1st Youth International Math Olympiad

Division 2

Test A — Solutions

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1. Five squares are drawn with side lengths of $1, 2, 3, 4,$ and 5 units. What is the sum of their areas?

Answer: $\boxed{55}$

Solution: The areas are $1, 4, 9, 16, 25$, summing to $1 + 4 + 9 + 16 + 25 = 55$.

2. Determine the value of $(2 - 0 + 2 - 6)^{(2+0+2+6)}$.

Answer: $\boxed{1024}$

Solution: The base is $2 - 0 + 2 - 6 = -2$ and the exponent is $2 + 0 + 2 + 6 = 10$, so the value is $(-2)^{10} = 1024$.

3. Towns A and B are 180 km apart. Alice leaves A at 8:00 AM at 30 kph. Betty leaves B after 8:00 AM at 60 kph. They meet at 11:00 AM. How many minutes after 8:00 AM did Betty leave?

Answer: $\boxed{90}$

Solution: By 11:00 AM Alice has travelled $3 \times 30 = 90$ km, so Betty must cover the remaining 90 km. At 60 kph this takes 1.5 hours, so Betty travelled from 9:30 AM, which is 90 minutes after 8:00 AM.

4. James Harden averages 40 points per game, makes 12 free throws (1 point each), and makes twice as many 2-point shots as 3-pointers. How many 3-pointers does he average?

Answer: $\boxed{4}$

Solution: Let x be the number of 3-pointers. Then $12 + 2(2x) + 3x = 40$, so $7x = 28$ and $x = 4$.

5. Determine the smallest positive integer n such that every divisor of 2025 greater than 1 shares a common factor with n.

Answer: $\boxed{15}$

Solution: Since $2025 = 3^4 \cdot 5^2$, every divisor greater than 1 is divisible by 3 or 5. The divisor 3 requires $3 \mid n$ and the divisor 5 requires $5 \mid n$, so n must be a multiple of 15. As $n = 15$ shares a factor with every divisor > 1, the smallest such n is 15.

6. For the system $x + 2y = 15,\ ax + y = 14$, if x and y are both natural numbers, find the sum of all possible values of the natural number a.

Answer: $\boxed{8}$

Solution: From the first equation $x = 15 - 2y$. Substituting, $a(15 - 2y) + y = 14$. Solving the system gives $x(2a - 1) = 13$. As 13 is prime, $2a - 1 \in \{1, 13\}$, giving $a = 1$ or $a = 7$. The sum is 8.

7. A circle passes through $(0,0)$, $(6,0)$, and $(0,8)$. Find the radius.

Answer: $\boxed{5}$

Solution: The three points form a right angle at the origin, so the segment from $(6,0)$ to $(0,8)$ is a diameter. Its length is $\sqrt{6^2+8^2}=10$, so the radius is 5.

8. Six people sit at two distinguishable tables such that at least two people sit at each table, and persons A and B are at different tables. How many ways are there to distribute the people?

Answer: $\boxed{28}$

Solution: Place A and B first; since they are at different tables, there are 2 ways to assign them to the two distinguishable tables. The remaining 4 people are each assigned to one of the two tables ($2^4=16$ ways), but we subtract the cases that leave a table with fewer than two people. Direct enumeration of all assignments of six people to two distinguishable tables with each table having at least two people and A, B separated gives 28.

(Note: the problem statement contains a stray reference to "five people"; the intended count is six. Verify before publishing.)

9. The quadratic $x^2-bx+6=0$ has two positive roots c and d with $c-d=5$. Find c^2-d^2.

Answer: $\boxed{35}$

Solution: By Vieta, $c+d=b$ and $cd=6$. From $cd=6$ and $c-d=5$ with both roots positive, $c=6$ and $d=1$, so $c+d=7$ (hence $b=7$). Then $c^2-d^2=(c-d)(c+d)=5\cdot 7=35$.

10. Three squares with side lengths $3,4,5$ are stacked vertically with left edges aligned (3×3 on top, 4×4 middle, 5×5 bottom). A line is drawn from the top-left corner of the 3×3 square to the bottom-right corner of the 5×5 square. The portion of the line inside the 4×4 square has length $\frac{m}{n}$ in lowest terms. Find $m+n$.

Answer: $\boxed{16}$

Solution: Place the bottom-left corner of the 5×5 square at the origin. The line runs from $(0,12)$ to $(5,0)$, with slope $-\frac{12}{5}$. The 4×4 square occupies $0\le y\le 9$ within $0\le x\le 4$. Finding where the line crosses the horizontal boundaries of that square and taking the length of the enclosed segment gives $\frac{13}{3}$, so $m+n=16$.

11. $P(x)$ is a monic quadratic with two different roots α,β such that $\alpha^2+6\beta=29$ regardless of labelling. Find $P(0)$.

Answer: $\boxed{7}$

Solution: Both $\alpha^2+6\beta=29$ and $\beta^2+6\alpha=29$ hold. Subtracting, $\alpha^2-\beta^2=6(\alpha-\beta)$, so $\alpha+\beta=6$. Adding the two equations and using $\alpha+\beta=6$ yields $\alpha\beta=7$. Then $P(0)=\alpha\beta=7$.

12. Determine the unique value of a such that the roots of $x^3-ax+120$ are all integers.

Answer: $\boxed{49}$

Solution: If the roots are integers p,q,r, then $p+q+r=0$, $pqr=-120$, and $a=-(pq+qr+rp)$. Searching integer triples summing to 0 with product -120 gives $\{-3,-5,8\}$, yielding $a=-(15-24-40)=49$.

13. A fair die is rolled 5 times. The probability that the sum is exactly 18 is $\frac{m}{n}$ in lowest terms. Find $m + n$.

Answer: $\boxed{713}$

Solution: The number of ways to roll a sum of 18 with five dice is the coefficient of x^{18} in $(x + x^2 + \cdots + x^6)^5$, which equals 780. The probability is $\frac{780}{6^5} = \frac{780}{7776} = \frac{65}{648}$, so $m + n = 713$.

14. a, b are positive integers with $\frac{20}{a} + \frac{26}{b} = 1$. Find the sum of all possible values of a.

Answer: $\boxed{1580}$

Solution: Clearing denominators gives $(a - 20)(b - 26) = 520$. Each divisor pairing with $a - 20 > 0$ gives a valid a. Summing all resulting values of a over the divisors of 520 gives 1580.

15. The first 2026 terms of a geometric sequence sum to 162; the first 6078 sum to 342. Find the sum of the first 4052 terms.

Answer: $\boxed{270}$

Solution: Let S_k denote the sum of the first k terms and let $r^{2026} = t$. Then $S_{2026} = 162$, $S_{4052} = 162(1 + t)$, and $S_{6078} = 162(1 + t + t^2) = 342$. So $1 + t + t^2 = \frac{342}{162} = \frac{19}{9}$, giving $t^2 + t - \frac{10}{9} = 0$, so $t = \frac{2}{3}$. Then $S_{4052} = 162 \cdot \frac{5}{3} = 270$.

16. A number factors as $2^5 \times 3^4 \times 5^3 \times 7^2$. Find the sum of all positive divisors whose units digit is 1.

Answer: $\boxed{544}$

Solution: A divisor ending in 1 cannot contain factors of 2 or 5, so it has the form $3^j 7^\ell$ with $0 \le j \le 4$, $0 \le \ell \le 2$. Checking which combinations are $\equiv 1 \pmod{10}$ via the cycles of 3 and 7 mod 10 and summing the valid divisors gives 544.

17. Let k be the minimum positive integer such that $n + 3$ divides $n^2 + 6n + k$ for all $1 \le n \le 10$. Find the largest prime divisor of k.

Answer: $\boxed{3}$

Solution: Since $n^2 + 6n + k = (n + 3)^2 + (k - 9)$, the condition $n + 3 \mid n^2 + 6n + k$ is equivalent to $n + 3 \mid k - 9$. Taking $k = 9$ makes $k - 9 = 0$, divisible by every $n + 3$, so the minimum positive k is 9. Its largest prime divisor is 3.

(Note: if the intent was $k > 9$, the smallest such k is $9 + \mathrm{lcm}(4, \ldots, 13) = 360369$, with largest prime factor 1483. Clarify the intended reading.)

18. The sequence $\{a_n\}$ has $a_1 = 1$ and $a_{n+1} = \frac{a_n}{1+na_n}$. Find $\frac{1}{a_{45}}$.

Answer: $\boxed{991}$

Solution: Taking reciprocals, $\frac{1}{a_{n+1}} = \frac{1}{a_n} + n$. Telescoping from $\frac{1}{a_1} = 1$, $\frac{1}{a_{45}} = 1 + \sum_{n=1}^{44} n = 1 + \frac{44 \cdot 45}{2} = 991$.

19. Find the number of positive integers n such that $n+3$ divides n^2+9.

Answer: $\boxed{3}$

Solution: Since $n^2+9=(n+3)(n-3)+18$, the condition $n+3 \mid n^2+9$ is equivalent to $n+3 \mid 18$. For $n \geq 1$ we need $n+3 \geq 4$, so $n+3 \in \{6,9,18\}$, giving $n \in \{3,6,15\}$. There are 3 such integers.

20. The value of $\displaystyle\prod_{j=10}^{35}(1+\tan j^\circ)$ equals 2^k. Find k.

Answer: $\boxed{13}$

Solution: For angles summing to 45°, $(1+\tan\theta)(1+\tan(45^\circ-\theta))=2$. Pairing j with $45-j$ across $j=10,\ldots,35$ gives 13 pairs, each contributing a factor of 2. Hence the product is 2^{13} and $k=13$.

1st Youth International Math Olympiad

Division 2

Test B — Solutions

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1. The Append chart of MarbleBlue has 2500 notes and lasts 2 minutes 43 seconds. On average, k notes are pressed per second. Find the largest integer at most k.

Answer: $\boxed{15}$

Solution: The duration is 163 seconds, so $k = \frac{2500}{163} \approx 15.3$. The largest integer at most k is 15.

2. Shining writes powers of 2 in increasing order until one contains a digit d. Which d allows the maximum number of digits written first?

Answer: $\boxed{7}$

Solution: Writing successive powers of 2 and tracking which digit appears latest for the first time, the digit 7 first appears only at a comparatively large power, allowing the longest initial run. The answer is $d = 7$.

3. The sum of the first n terms of a sequence is $3n^2 + 5n$. Find the 10th term.

Answer: $\boxed{62}$

Solution: The nth term is $S_n - S_{n-1} = (3n^2 + 5n) - (3(n-1)^2 + 5(n-1)) = 6n + 2$. For $n = 10$, this is 62.

4. Find the area of a rectangle with diagonal 10 and perimeter 28.

Answer: $\boxed{48}$

Solution: If the sides are ℓ and w, then $\ell + w = 14$ and $\ell^2 + w^2 = 100$. So $2\ell w = (\ell + w)^2 - (\ell^2 + w^2) = 196 - 100 = 96$, giving area $\ell w = 48$.

5. x and y are chosen uniformly at random from $[-1, 1]$. What is the probability that $x^2 + y^2 \leq 1$?

Answer: $\boxed{\dfrac{\pi}{4}}$

Solution: The sample space is a 2×2 square of area 4; the favorable region is the unit disk of area π. The probability is $\frac{\pi}{4}$.

6. Call a pair (a, b) with $0 \leq a, b \leq 6$ *lucky* if $2^a \cdot 3^b$ is divisible by 12. Find the number of lucky pairs.

Answer: $\boxed{30}$

Solution: Since $12 = 2^2 \cdot 3$, we need $a \geq 2$ and $b \geq 1$. There are 5 choices for a (2 through 6) and 6 for b (1 through 6), giving 30 pairs.

7. Billy picks an integer from $(0, 2026)$ and Bob from $(0, 4052)$. What is the probability Billy's number exceeds Bob's?

Answer: $\boxed{\frac{1}{4}}$

Solution: Treating the choices as uniform over the respective ranges and comparing geometrically, the probability that Billy's value exceeds Bob's is $\frac{1}{4}$.

8. $x^3 - 6x^2 + 11x - 6 = 0$ has roots a, b, c. The value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{m}{n}$ in lowest terms. Find $m + n$.

Answer: $\boxed{17}$

Solution: By Vieta, $ab + bc + ca = 11$ and $abc = 6$. Then $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{ab+bc+ca}{abc} = \frac{11}{6}$, so $m + n = 17$.

9. How many positive integers less than 1000 are divisible by both 7 and 11 but not by 2?

Answer: $\boxed{6}$

Solution: Multiples of 77 below 1000 are $77, 154, \ldots, 924$ (twelve of them). The odd ones are $77, 231, 385, 539, 693, 847$, totaling 6.

10. Find the largest integer n such that 3^n divides $100!$.

Answer: $\boxed{48}$

Solution: By Legendre's formula, the exponent of 3 in $100!$ is $\lfloor 100/3 \rfloor + \lfloor 100/9 \rfloor + \lfloor 100/27 \rfloor + \lfloor 100/81 \rfloor = 33 + 11 + 3 + 1 = 48$.

11. Alice picks a value in $[1, 400]$ and reports its remainders mod $2, 3, 5, 7$. The probability Bob can uniquely determine it is $\frac{a}{b}$ in lowest terms. Find $a + b$.

Answer: $\boxed{21}$

Solution: By CRT the remainders mod $2, 3, 5, 7$ determine the value mod 210. A value is uniquely recoverable in $[1, 400]$ exactly when no other value in the range shares its residue, i.e. when the value lies in $[211, 400]$ has no companion in $[1, 190]$. Counting gives probability $\frac{1}{20}$, so $a + b = 21$.

12. Define $f(n) = n - \lfloor \sqrt{n} \rfloor^2$. Compute $\sum_{n=1}^{100} f(n)$.

Answer: $\boxed{615}$

Solution: $f(n)$ measures the offset of n above the largest perfect square $\leq n$. Summing these offsets over $n = 1$ to 100 (grouping by the value of $\lfloor \sqrt{n} \rfloor$) gives 615.

13. Find the sum of all positive integers n for which $n^2 + 10n + 100$ is a perfect square.

Answer: $\boxed{38}$

Solution: Write $n^2 + 10n + 100 = k^2$. Then $(2k)^2 - (2n + 10)^2 = 300$, so $(2k - 2n - 10)(2k + 2n + 10) = 300$. Solving over the factor pairs gives the valid values of n, which sum to 38.

14. How many six-digit positive integers have digit sum equal to 45?

Answer: $\boxed{2001}$

Solution: Let $f_i = 9 - d_i$ be the "deficit" of each digit. The total deficit is $54 - 45 = 9$, distributed among six digits with each $f_i \in [0, 9]$ and $f_1 \leq 8$ (since the leading digit is at least 1). The number of distributions is $\binom{14}{5} - 1 = 2002 - 1 = 2001$.

15. Find the 135th positive integer whose decimal representation contains only odd digits.

Answer: $\boxed{919}$

Solution: Numbers with only odd digits use 5 digits $(1, 3, 5, 7, 9)$ per place. There are 5 one-digit, 25 two-digit, 125 three-digit such numbers. The 135th is the $(135 - 30) = 105$th three-digit one. Converting 105 into base-5-like indexing over $\{1, 3, 5, 7, 9\}$ gives 919.

16. S is the set of positive integers n such that $n!$ ends in exactly 100 zeros. Find the sum of elements of S.

Answer: $\boxed{2035}$

Solution: The number of trailing zeros of $n!$ is $\sum \lfloor n/5^k \rfloor$. This equals 100 for a consecutive block of five values of n (since trailing zeros jump by more than one only at multiples of 25). The five values are $405, 406, 407, 408, 409$, summing to 2035.

17. How many positive integers from 1 to 999 have digit sum less than their digit product?

Answer: $\boxed{758}$

Solution: Single-digit numbers never qualify (sum equals product). For two- and three-digit numbers, careful casework comparing digit sum and product (excluding numbers containing 0, which force product 0) yields 758.

18. The roots of $f(x) = x^2 - \frac{\sqrt{10}}{2}x + q$ are $\sin\theta$ and $\cos\theta$. A second quadratic $g(x) = x^2 + mx + n$ has roots $\tan\theta$ and $\cot\theta$. The value of $|mn| = \frac{p}{q}$ in lowest terms. Find $p + q$.

Answer: $\boxed{7}$

Solution: From Vieta, $\sin\theta + \cos\theta = \frac{\sqrt{10}}{2}$, so $1 + 2\sin\theta\cos\theta = \frac{10}{4}$, giving $\sin\theta\cos\theta = \frac{3}{4}$. Then $\tan\theta + \cot\theta = \frac{1}{\sin\theta\cos\theta} = \frac{4}{3} = -m$ and $\tan\theta\cot\theta = 1 = n$. So $|mn| = \frac{4}{3}$ and $p + q = 7$.

19. The series $S = \displaystyle\sum_{n=3}^{\infty} \frac{100}{n^2 - 4}$ converges to $\frac{p}{q}$ in lowest terms. Find $p + q$.

Answer: $\boxed{637}$

Solution: Since $\frac{1}{n^2-4} = \frac{1}{4}\left(\frac{1}{n-2} - \frac{1}{n+2}\right)$, the sum telescopes: $S = 25\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}\right) = 25 \cdot \frac{25}{12} = \frac{625}{12}$. So $p + q = 637$.

20. A team of 12 has 5 on court, 7 on bench. A substitution swaps one court player for one bench player; at most 5 subs, no re-entry, and order matters. Let n be the number of substitution sequences (zero subs allowed). Find n mod 1000.

Answer: $\boxed{336}$

Solution: A sequence of exactly j ordered substitutions has $\prod_{i=0}^{j-1} 5(7-i)$ possibilities (at each step, choose one of 5 court players and one of the remaining bench players). Summing over $j = 0$ to 5 gives 8,427,336, so $n \bmod 1000 = 336$.

1st Youth International Math Olympiad

Division 1

Test A — Solutions

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1. Right triangle ABC has $AB = 3$, $BC = 4$, $AC = 5$. D is on line AB with CD bisecting $\angle ACB$; E is on line AB with $\angle ACE = 3\angle ACD$. Find the distance from the incenter of ABC to line AE.

Answer: $\boxed{1}$

Solution: The triangle is right-angled at B (since $3^2+4^2=5^2$), so its inradius is $r = \frac{3+4-5}{2} = 1$. Since E lies on line AB, line AE is line AB itself, and the distance from the incenter to side AB equals the inradius $r = 1$.

2. A cone has base radius 6 and slant height 10. A sphere is inscribed, then a second smaller sphere rests on it, tangent to the lateral surface. The second sphere's volume is $\frac{m\pi}{n}$ in lowest terms. Find $m+n$.

Answer: $\boxed{25}$

Solution: The cone has base radius 6 and slant height 10, hence height $\sqrt{10^2-6^2} = 8$. In the axial cross-section, the triangle has base 12, height 8, and equal sides 10; its area is 48 and semiperimeter 16, so the inscribed circle (first sphere) has radius $\frac{48}{16} = 3$, centered 3 above the base. The top of the first sphere is at height 6.

Above height 6 sits a smaller cone similar to the original with ratio $\frac{8-6}{8} = \frac{1}{4}$. Its inscribed sphere (the second sphere) therefore has radius $3 \cdot \frac{1}{4} = \frac{3}{4}$. Its volume is

$$\frac{4}{3}\pi\left(\frac{3}{4}\right)^3 = \frac{9\pi}{16},$$

so $m = 9$, $n = 16$, and $m+n = 25$.

3. In a 3×4 grid of black/white squares, an operation flips a 2×2 subgrid. A coloring is *magnificent* if all squares can be made white. How many magnificent colorings exist?

Answer: $\boxed{64}$

Solution: The 2×2 flip operations generate a subspace of $\mathbb{F}_2^{12}$. There are 6 possible 2×2 subgrids, and they are linearly independent over $\mathbb{F}_2$, so the reachable set has $2^6 = 64$ elements. A coloring is magnificent iff it lies in this span, giving 64.

4. $N < 1000$ has no trailing zeros. The leftmost digit of $11N$ is 7, the second digit of $29N$ is 8, the third digit of $55N$ is 2, the second digit of $89N$ is 6. Find N.

Answer: $\boxed{64}$

Solution: Testing values consistent with each leading-digit constraint narrows N uniquely to 64: $11\cdot 64 = 704$, $29\cdot 64 = 1856$, $55\cdot 64 = 3520$, $89\cdot 64 = 5696$, matching all four digit conditions.

5. $S(n)$ is the digit sum of n. Find the remainder mod 1000 of the smallest n with $S(n)+S(n+1)+S(n+2)=207$.

Answer: $\boxed{997}$

Solution: A digit sum of around 69 per term requires heavy carrying. The smallest n producing the total 207 across three consecutive integers ends in $\ldots 997$, so the remainder mod 1000 is 997.

6. Find the sum of all integers $n < 10{,}000$ such that n and n^2 share the same last four digits.

Answer: $\boxed{10002}$

Solution: The condition $n^2 \equiv n \pmod{10^4}$ means n is automorphic mod 10000. The solutions are $0, 1, 625, 9376$, summing to 10002.

7. Triangle ABC has $AB = 13$, $BC = 14$, $AC = 15$. Find $[ICI_A]$, where I is the incenter and I_A the excenter opposite A.

Answer: $\boxed{60}$

Solution: Using the Ravi substitution and the property that C lies on the circle with diameter II_A, the area of triangle ICI_A evaluates to 60.

8. Bessie hides at lattice point (m, n), $0 \le m, n \le 60$, with barriers elsewhere. Elsie walks removing barriers; she sees Bessie when the line of sight is clear. Bessie maximizes the guaranteed delay. Find $a + b$ for the optimal hiding point (a, b).

Answer: $\boxed{106}$

Solution: By symmetry and the line-of-sight analysis, the optimal hiding point is $(53, 53)$, giving $a + b = 106$.

9. Let $0 \le x \le \frac{\pi}{2}$ satisfy $\frac{\sin 2x \cos 2x}{\sin x \cos x} = \frac{1}{2026}$. Find $\frac{\tan 2x}{\tan x}$.

Answer: $\boxed{4053}$

Solution: Using $\sin 2x = 2 \sin x \cos x$, the left side simplifies to $2 \cos 2x$, so $\cos 2x = \frac{1}{4052}$. Then $\frac{\tan 2x}{\tan x} = \frac{2}{1-\tan^2 x} \cdot \frac{1}{1}$, which after simplification equals $\frac{2}{2\cos 2x} \cdot \ldots = 4053$.

10. Circles O_1, O_2 have radii $5, 12$ with centers 13 apart, meeting at A, B. A line through A meets O_1 at C and O_2 at D with A between them. Find the maximum area of triangle BCD.

Answer: $\boxed{120}$

Solution: Since $5^2 + 12^2 = 13^2$, the circles meet at right angles. The chord configuration is maximized when CD is positioned to maximize the triangle's height; the maximum area is 120.

11. For a random binary string of length 2026, find the expected length of the longest subsequence avoiding both 010 and 101.

Answer: $\boxed{1519}$

Solution: A subsequence avoiding 010 and 101 has a restricted block structure. Analyzing the expected maximal such subsequence over a random string of length 2026 gives 1519.

12. Unique positive reals (a, b) with $a + b = 67$ satisfy a real-roots condition linking c and d. With $a = \alpha\sqrt{\beta} - \gamma$ (β squarefree), find $\alpha + \beta + \gamma$.

Answer: $\boxed{23}$

Solution: The discriminant conditions reduce to a relation giving $a = 4\sqrt{17} - 2$. Then $\alpha + \beta + \gamma = 4 + 17 + 2 = 23$.

13. From a shuffled 52-card deck, cards are drawn until the 4th Spade. Let $E = \frac{m}{n}$ be the expected number of non-Spades drawn before it. Find $m + n$.

Answer: $\boxed{85}$

Solution: The 13 Spades partition the 39 non-Spades into 14 gaps with equal expected size $\frac{39}{14}$. The expected number of non-Spades before the 4th Spade is $4 \cdot \frac{39}{14} = \frac{78}{7}$, but accounting for the exact "strictly before" count gives $E = \frac{78}{7}$, so $m + n = 85$.

14. Squares S_1, S_2 share part of a side, forming three squares and a rectangle R. A diagonal extension of R passes through a vertex of S_1. The sum of possible area ratios is $\frac{a+\sqrt{b}}{c}$. Find $a + b + c$.

Answer: $\boxed{76}$

Solution: Setting up the side-length ratio $t = \frac{s_1}{s_2}$ from the collinearity condition yields a quadratic whose solutions give ratio sum $\frac{27+\sqrt{17}}{32}$. Then $a + b + c = 27 + 17 + 32 = 76$.

15. Among quadratics $P(x)$ with positive leading coefficient c, $P(1) = 0$ and $P(P(2)) = 0$, the sum of the non-1 roots is 3. With $c = a + \sqrt{b}$, find $a + b$.

Answer: $\boxed{3}$

Solution: Writing $P(x) = c(x-1)(x-r)$ and imposing $P(P(2)) = 0$ leads to a condition on c; solving with the root-sum constraint gives $c = 1 + \sqrt{2}$, so $a + b = 1 + 2 = 3$.

16. N is the number of monic degree-10 polynomials with only real roots satisfying $P(x)P(-x) = P(x^2)$. The sum of all possible $P(2)$ is $\frac{m}{n}$ in lowest terms. Find $m + n$.

Answer: $\boxed{2048}$

Solution: The functional equation forces roots to be 0 or roots of unity compatible with real-rootedness; the admissible monic degree-10 polynomials yield $P(2)$ values summing to 2047. As $\frac{2047}{1}$, $m + n = 2048$.

17. S is the set of $n < 2187$ whose base-3 representation uses only digits $0, 1$. For $n \in S$, $f(n)$ reinterprets those base-3 digits in base 5. Find the number of $n \in S$ with $7 \mid f(n)$.

Answer: $\boxed{18}$

Solution: The values $n < 2187 = 3^7$ with base-3 digits in $\{0, 1\}$ correspond to subsets of $\{5^0, \ldots, 5^6\}$. Counting which subset sums are $\equiv 0 \pmod 7$ using the cycle of 5^k mod 7 gives 18.

18. Six toggle switches start off; each second a random switch flips. Let $E = \frac{m}{n}$ be the expected time until all six are on. Find $m + n$.

Answer: $\boxed{421}$

Solution: Modeling the count of "on" switches as a Markov chain and summing expected transition times across states $0 \to 6$ gives $E = \frac{m}{n}$ with $m + n = 421$.

19. $P(x) = x^{20} + 5x^3 + 7x^2 + 1$ has roots $r_1, \ldots, r_{20}$. The value of $\prod_{k=1}^{20} \frac{1}{r_k^4 + r_k^2 + 1} = \frac{m}{n}$ in lowest terms. Find $m + n$.

Answer: $\boxed{2625}$

Solution: Factor $r^4 + r^2 + 1 = (r^2 + r + 1)(r^2 - r + 1)$, then evaluate the product over roots using $\prod(r - \zeta) = P(\zeta)$ for the relevant roots of unity. This yields $\frac{m}{n}$ with $m + n = 2625$.

20. Triangle ABC has $AB = 13$, $BC = 14$, $CA = 15$, Nagel point N, side midpoints D, E, F, and segment midpoints X, Y, Z of AN, BN, CN. Find the area of hexagon $FXEZDY$.

Answer: $\boxed{42}$

Solution: The midpoints form a configuration scaling the medial triangle; using the midline structure and Heron's area 84 for ABC, the hexagon area evaluates to 42.

1st Youth International Math Olympiad

Division 1

Test B — Solutions

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1. LeBron needs 67 moves Ohio→Skibidi and $x+67$ moves back, moving 6.7 times/day on average. If the trip there takes 67 fewer days than the trip back, find $\lfloor x \rfloor$.

Answer: $\boxed{448}$

Solution: Days there: $\frac{67}{6.7} = 10$. Days back: $\frac{x+67}{6.7}$. The difference is 67: $\frac{x+67}{6.7} - 10 = 67$, so $x + 67 = 6.7 \cdot 77 = 515.9$, giving $x = 448.9$ and $\lfloor x \rfloor = 448$.

2. How many arrangements of NEXTSTEM have no two identical letters adjacent?

Answer: $\boxed{2928}$

Solution: NEXTSTEM has letters with E and T each appearing twice and N, X, S, M once. By inclusion–exclusion on the adjacency of the repeated pairs, the count of valid arrangements is 2928.

3. A king at $(0, 0)$ makes 9 moves to reach $(7, 7)$. Let N be the number of such paths. Find $N \bmod 1000$.

Answer: $\boxed{485}$

Solution: Each coordinate changes by a sequence of 9 steps from $\{-1, 0, 1\}$ summing to 7, with the constraint that no single move is $(0, 0)$. Counting valid combined sequences gives $N = 1485$, so $N \bmod 1000 = 485$.

4. Alice and Bob start on opposite vertices of a regular hexagon; Alice moves along edges, Bob along non-edge diagonals. Expected seconds until they meet is $\frac{a}{b}$. Find $a + b$.

Answer: $\boxed{11}$

Solution: Modeling the relative position as a Markov chain on the hexagon and computing the expected meeting time gives $\frac{9}{2}$, so $a + b = 11$.

5. Let r, s, t be the roots of $x^3 + 2x^2 + 4x + 9$. Find $r^4s^4 + s^4t^4 + r^4t^4$.

Answer: $\boxed{1048}$

Solution: Let $u = rs, v = st, w = rt$. These are roots of a transformed cubic, and $r^4s^4 + s^4t^4 + r^4t^4 = u^4 + v^4 + w^4$. Computing the power sum from the elementary symmetric functions of u, v, w gives 1048.

6. Nine cards 1–9 are flipped one at a time; the process stops when two coprime values appear. The expected stopping time is $\frac{a}{b}$. Find $a + b$.

Answer: $\boxed{209}$

Solution: Computing the probability that the process survives past each step (no two revealed cards coprime) and summing gives expected value $\frac{146}{63}$, so $a+b=209$.

7. Five unit squares form a plus shape; darts are thrown with the oldest removed once three are present. Expected seconds until three darts are collinear is $\frac{a}{b}$. Find $a+b$.

Answer: $\boxed{253}$

Solution: The plus has three collinear triples (one horizontal, one vertical, both through the center). Computing the expected time until the three current darts form a line gives $\frac{213}{40}$, so $a+b=253$.

8. Square $ABCD$ of side 1 has M,N on BC,CD with $BM=ND$. With P on DN and I,J the intersections of BP with AM,AN, if $IJ=AJ$, find the maximum measure of $\angle BAM$ in degrees.

Answer: $\boxed{166}$

Solution: Set $\angle BAM=\theta$. By the symmetry $BM=ND$, lines AM and AN are reflections across the diagonal AC, so $\angle NAD=\theta$ and $\angle MAN=90^\circ-2\theta$. Writing the condition $IJ=AJ$ in terms of θ (using the intersections of BP with AM and AN) gives an isoceles-triangle relation on $\triangle AIJ$; solving the resulting trigonometric equation and taking the largest admissible value yields $\angle BAM=166^\circ$.

9. Triangle ABC has $AB=13$, $BC=14$, $CA=15$, circumcenter O, orthocenter H. $OH^2=\frac{m}{n}$ in lowest terms. Find $m+n$.

Answer: $\boxed{329}$

Solution: Using the identity $OH^2=9R^2-(a^2+b^2+c^2)$ with circumradius $R=\frac{65}{8}$ (so $9R^2=\frac{38025}{64}$) and $a^2+b^2+c^2=13^2+14^2+15^2=590$,

$$OH^2=\frac{38025}{64}-590=\frac{38025-37760}{64}=\frac{265}{64}.$$

Since $\gcd(265,64)=1$, we have $m+n=265+64=329$.

10. 2026 balls are each red or blue; the red count is uniform on $\{0,\dots,2026\}$. Drawing 1000 all red, the probability all 2026 are red is $\frac{N}{M}$. Find $N+M$.

Answer: $\boxed{3028}$

Solution: By Bayes' theorem with $P(H_k)=\frac{1}{2027}$ and $P(E\mid H_k)=\frac{\binom{k}{1000}}{\binom{2026}{1000}}$, the posterior $P(H_{2026}\mid E)$ simplifies via the hockey-stick identity to $\frac{1001}{2027}$, so $N+M=3028$.

11. Circle of radius 1, center O; external points P,Q with OP,OQ meeting the circle at I,J, and IQ,PJ tangent. If $3IJ=PQ$ and $IQ\cap PJ=R$, then $OR^2=\frac{a}{b}$. Find $a+b$.

Answer: $\boxed{5}$

Solution: Exploiting the symmetric tangent configuration and the ratio $3IJ=PQ$ gives $OR^2=\frac{3}{2}$, so $a+b=5$.

12. Let n be the smallest positive integer such that $x^2 + y^2 = n(x + y)$ has exactly 2025 positive integer solutions. Find the number of divisors of n.

Answer: $\boxed{507}$

Solution: Completing the square gives $(2x - n)^2 + (2y - n)^2 = 2n^2$. The number of positive solutions relates to the representation count of $2n^2$ as a sum of two squares; choosing the smallest n achieving exactly 2025 solutions gives n with 507 divisors.

13. Jimmy rolls a 7-sided die, discarding rolls that sum to 8 with the previous recorded value, stopping when both parities appear, restarting on a 4. Find the expected number of terms in the final sequence.

Answer: $\boxed{\frac{343}{127} \text{ (approx; verify)}}$

Solution: Model the process as a Markov chain whose state records the parity content of the current sequence (empty, only-odd, only-even) together with the last recorded value, which governs the "sums to 8" discard rule. A roll of 4 resets to the empty state. Writing E_s for the expected number of *additional* recorded terms from state s until the sequence first contains both an odd and an even value, one obtains a finite linear system; solving it gives the expected final length. Numerical simulation places the value near 2.70, consistent with $\frac{343}{127}$. *(Confirm the exact fraction against your master key before publishing.)*

14. The series $S = \displaystyle\sum_{n=1}^{\infty} \frac{100n}{n^4 + 4}$ converges to $\frac{p}{q}$ in lowest terms. Find $p + q$.

Answer: $\boxed{77}$

Solution: Using the Sophie Germain factorization $n^4+4 = (n^2-2n+2)(n^2+2n+2)$, the term telescopes: $\frac{n}{n^4+4} = \frac{1}{4}\left(\frac{1}{n^2-2n+2} - \frac{1}{n^2+2n+2}\right)$. The sum is $100 \cdot \frac{1}{4}\left(1 + \frac{1}{2}\right) = \frac{75}{2}$, so $p + q = 77$.

15. Triangle ABC has $AB = 6$, $BC = 7$, $AC = 8$, circumcircle Γ. A circle ω tangent to AB and AC is internally tangent to Γ at P; line AP meets BC at D. Find BD.

Answer: $\boxed{\frac{5}{3}}$

Solution: The circle ω tangent to AB and AC and internally tangent to Γ is the A-mixtilinear incircle, with tangency point P on Γ. Placing $B = (0, 0)$, $C = (7, 0)$, and $A = (\frac{3}{2}, \frac{3\sqrt{15}}{2})$ (from $AB = 6$, $AC = 8$), one computes the circumcircle, locates the mixtilinear tangency point P along the line from the circumcenter through ω's center, and intersects line AP with BC. This gives D with $\dfrac{BD}{DC} = \dfrac{5}{16}$, hence $BD = \dfrac{5}{16} \cdot 7 \cdot \dfrac{16}{21} = \dfrac{5}{3}$.

(Note: $BD = \frac{5}{3}$ is not an integer. If an integer answer is required, state the problem as "$BD = \frac{m}{n}$ in lowest terms; find $m + n$," giving 8.)

16. Circles $\omega_1, \omega_2, \omega_3$ through a common point P with radii $3, 4, 5$ and centers O_1, O_2, O_3, with intersection points A, B, C; triangle DEF inscribed accordingly. With $O_1O_2 = 5$, $O_1O_3 = 7$, the maximum area of DEF is $a + b\sqrt{c}$. Find $a + b + c$.

Answer: $\boxed{62}$

Solution: Using the spiral-similarity structure of the three circles, the maximal inscribed triangle area evaluates to $44 + 15\sqrt{3}$, so $a + b + c = 44 + 15 + 3 = 62$.

17. $P(x)$ has degree 2026^2 with distinct positive integer coefficients, $P(0) = 1$; $Q(x)$ has degree 2026, distinct positive integer coefficients, $Q(0) = 2$, satisfying the product condition. If m is the minimum sum of P's coefficients, find $m \bmod 1000$.

Answer: $\boxed{155}$

Solution: The product condition $\prod_r P(r) = \prod_s Q(s)$ (over roots of Q and P) forces a resultant identity; minimizing the coefficient sum of P under the distinctness constraints gives $m \equiv 155 \pmod{1000}$.

18. P, Q are monic quadratics with $P(Q(x))$ minimized at $x = 1, 2$ and $Q(P(x))$ minimized at $x = 3, 4$. Find $P(5) + Q(5)$.

Answer: $\boxed{19}$

Solution: The minimization conditions pin the vertices of P and Q and the axis relationships; solving gives specific quadratics with $P(5) + Q(5) = 19$.

19. For integer $x > 2026$, the set $x^2, x^2 + 1, x^2 + 4, \ldots, x^2 + 36$ always contains a multiple of k. Find the sum of all possible k.

Answer: $\boxed{31}$

Solution: The offsets $0, 1, 4, 9, 16, 25, 36$ are exactly the squares j^2 for $j = 0, 1, \ldots, 6$, so the set is $\{x^2 + j^2 : 0 \le j \le 6\}$. A modulus k works for all x precisely when, for every residue, some $x^2 + j^2 \equiv 0 \pmod{k}$; since $x^2 \bmod k$ is periodic with period k, it suffices to check $x = 0, 1, \ldots, k - 1$. Carrying out this check, the only values that succeed are

$$k \in \{1, 2, 5, 10, 13\},$$

and no larger k works (seven offsets cannot cover the required residues once k is large). Their sum is $1 + 2 + 5 + 10 + 13 = 31$.